

SAFETY DATA SHEET – METHYLDIETHANOLAMINE (MDEA) SOLUTION

1. Identification

- Product: Aqueous MDEA (N-methyldiethanolamine, CAS 105-59-9) solution, used in amine treating / regeneration (V-501, E-501, P-501).
- Typical lean amine strength in service: 45–50 wt% MDEA.

2. Hazards

- Causes serious eye irritation; may cause skin irritation.
- Rich amine may release hydrogen sulphide (H₂S) – highly toxic, flammable gas.
- Hot amine can cause burns.

3. Operating limits relevant to safety

- Rich amine H₂S loading limit: ≤ 0.45 mol H₂S / mol amine (corrosion and H₂S release risk above this).
- Lean amine residual loading target: < 0.01 mol/mol.

4. Exposure limits (as commonly cited)

- H₂S: TLV 1 ppm (8-h TWA), 5 ppm STEL (ACGIH). IDLH 50 ppm.
- Personal H₂S monitor alarm set 5 ppm (low) / 10 ppm (high).

5. PPE and handling

- Chemical goggles + face shield, nitrile / PVC gloves, chemical-resistant apron.
- Personal H₂S monitor mandatory in amine area; SCBA for entry where H₂S > 10 ppm.
- Sample rich amine only through closed-loop sampler; buddy system.

6. First aid

- Eyes: flush with water for at least 15 minutes, seek medical attention.
- Skin: remove contaminated clothing, wash with plenty of water.
- Inhalation (H₂S): move to fresh air only if rescuer protected with SCBA; give oxygen; CPR if required.
- Ingestion: do not induce vomiting; rinse mouth; seek medical help.

7. Spill / fire

- Contain with sand/earth, prevent entry to drains. MDEA is combustible (flash point ~ 127 degC neat).
- Extinguish with water spray, foam, dry chemical or CO₂.